PRACTICAL NO. 1

Retail Data Warehouse — Analytics Assignment

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Roll No: [DT-22007]

Course: Data Analytics and Visualization (DAV)

Dataset Schema

The dataset simulates a national retail company's monthly transaction data warehouse. It contains 1,848 records spanning 2 years (2023–2024) across 5 regions, 7 product categories, and multiple provinces.

Table: RETAIL_TRANSACTIONS

Column Name	Data Type	Description
YEAR	INTEGER	Transaction year (2023 or 2024)
MONTH	INTEGER	Transaction month (1–12)
REGION	VARCHAR	Sales region (North, South, East, West, Central)
STATE_PROVINCE	VARCHAR	Province (Punjab, Sindh, KPK, Balochistan, AJK, GB)
CATEGORY	VARCHAR	Product category (Electronics, Clothing, etc.)
SALES	DECIMAL(12,2)	Total sales revenue in PKR
UNITS	INTEGER	Total number of units sold

Sample Data Overview

Metric	Value
Total Records	1,848
Time Period	Jan 2023 – Dec 2024
Regions	5 (North, South, East, West, Central)
Product Categories	7
Provinces	6
Total Sales (2023+2024)	PKR 149.1M (approx.)

Tasks 1–10: Core Analytics

Task 1: Regional Sales Performance

Find total sales by region, ordered from highest to lowest, to identify top-performing geographical areas.

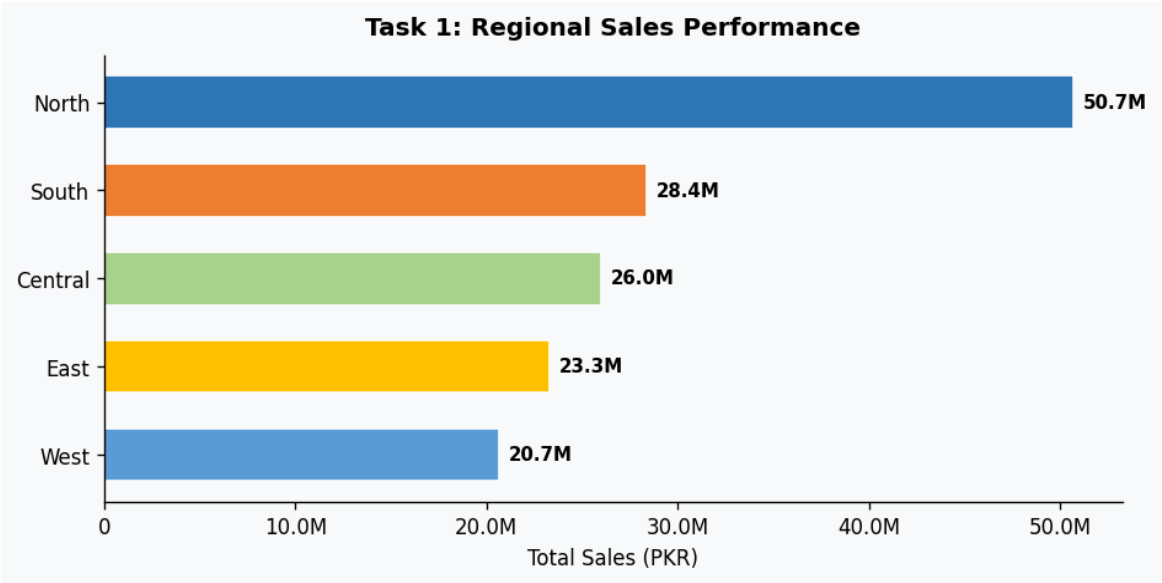
SQL Query

```
SELECT REGION,
       SUM(SALES) AS TOTAL_SALES
FROM   RETAIL_TRANSACTIONS
GROUP  BY REGION
ORDER  BY TOTAL_SALES DESC;
```

Query Results

#	Region	Total Sales
1	North	PKR 50.73M
2	South	PKR 28.40M
3	Central	PKR 25.98M
4	East	PKR 23.26M
5	West	PKR 20.68M

Visualization



Task 2: Category Sales Performance

Find total sales by product category to understand which categories drive the most revenue.

SQL Query

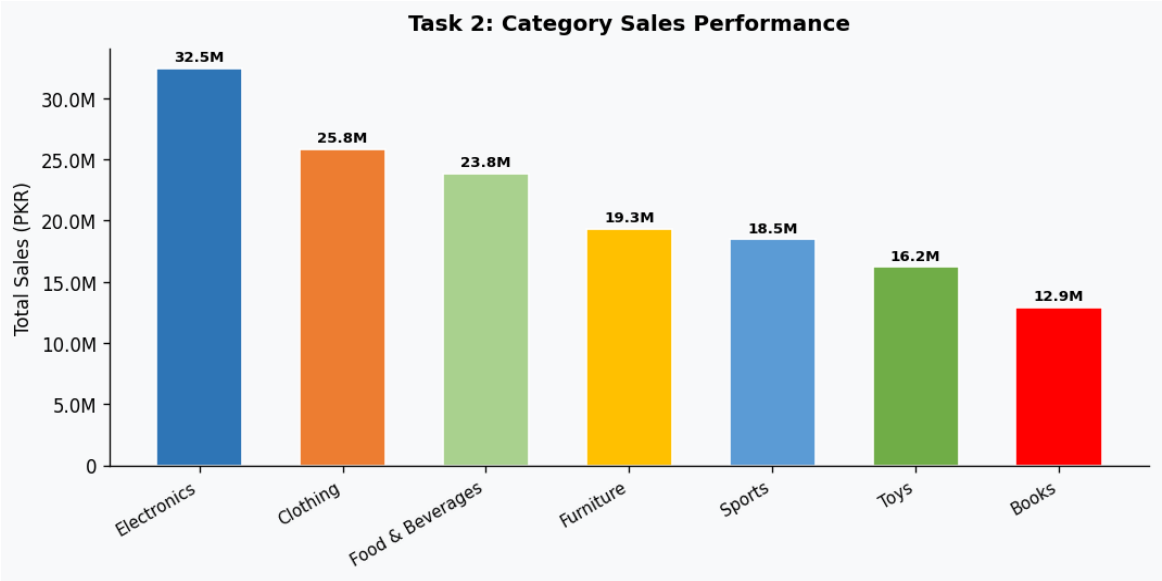
```
SELECT CATEGORY,
       SUM(SALES) AS TOTAL_SALES
FROM   RETAIL_TRANSACTIONS
GROUP  BY CATEGORY
ORDER  BY TOTAL_SALES DESC;
```

Query Results

#	Category	Total Sales
1	Electronics	PKR 32.46M
2	Clothing	PKR 25.83M

3	Food & Beverages	PKR 23.84M
4	Furniture	PKR 19.34M
5	Sports	PKR 18.48M
6	Toys	PKR 16.20M
7	Books	PKR 12.89M

Visualization



Task 3: Monthly Sales Trend

Calculate total sales per month across all years to observe seasonal and growth trends over time.

SQL Query

```
SELECT YEAR, MONTH,
       SUM(SALES) AS TOTAL_SALES
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, MONTH
ORDER  BY YEAR, MONTH;
```

Query Results

Year-Month	Total Sales
2023-01	PKR 4.77M
2023-02	PKR 4.43M
2023-03	PKR 5.14M
2023-04	PKR 5.35M

2023-05	PKR 5.77M
2023-06	PKR 6.10M

Visualization

Task 4: Monthly Units Trend

Calculate total units sold per month to analyze volume trends independent of pricing effects.

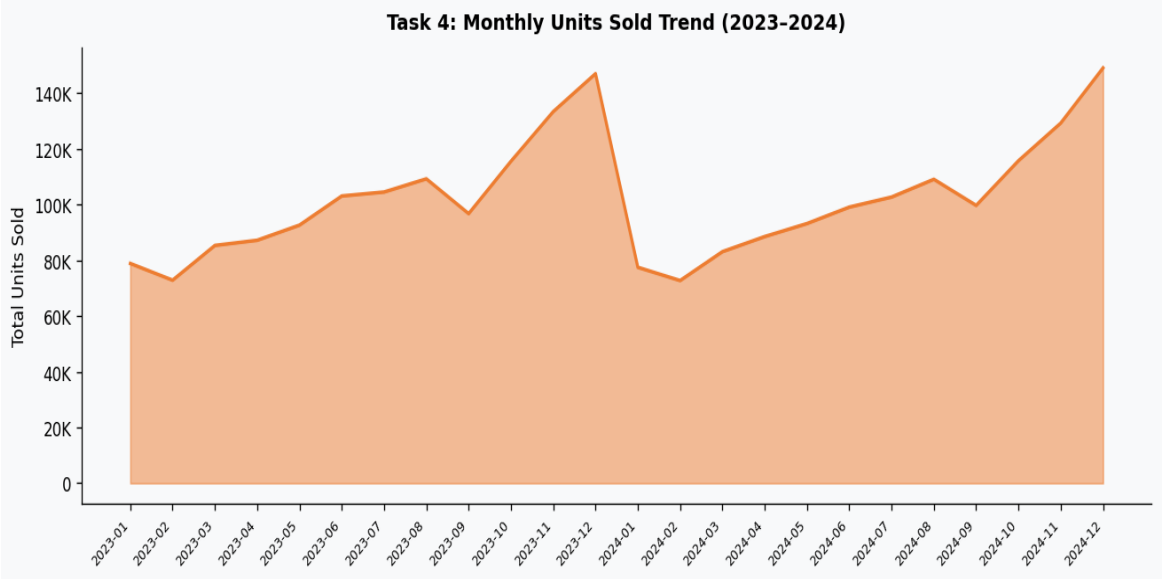
SQL Query

```
SELECT YEAR, MONTH,
       SUM(UNITS) AS TOTAL_UNITS
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, MONTH
ORDER  BY YEAR, MONTH;
```

Query Results

Year-Month	Total Units
2023-01	78,899
2023-02	72,893
2023-03	85,363
2023-04	87,227
2023-05	92,660
2023-06	103,123

Visualization



Task 5: Region vs Category Sales (Stacked)

Break down sales by both region and category to identify cross-dimensional performance patterns.

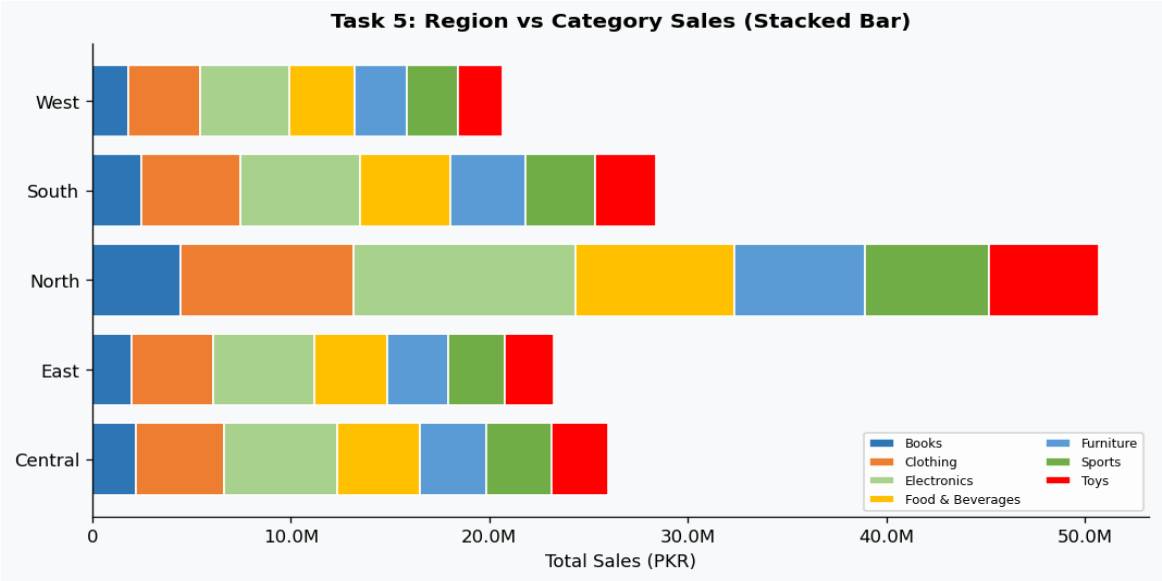
SQL Query

```
SELECT REGION, CATEGORY,
       SUM(SALES) AS TOTAL_SALES
FROM   RETAIL_TRANSACTIONS
GROUP  BY REGION, CATEGORY
ORDER  BY REGION, TOTAL_SALES DESC;
```

Query Results

Region	Category	Total Sales
Central	Books	PKR 2.21M
Central	Clothing	PKR 4.45M
Central	Electronics	PKR 5.65M
Central	Food & Beverages	PKR 4.20M
Central	Furniture	PKR 3.35M
Central	Sports	PKR 3.27M
Central	Toys	PKR 2.86M
East	Books	PKR 1.99M

Visualization



Task 6: Month vs Category Sales (Stacked)

Examine how each product category performs across the twelve months to detect seasonal category patterns.

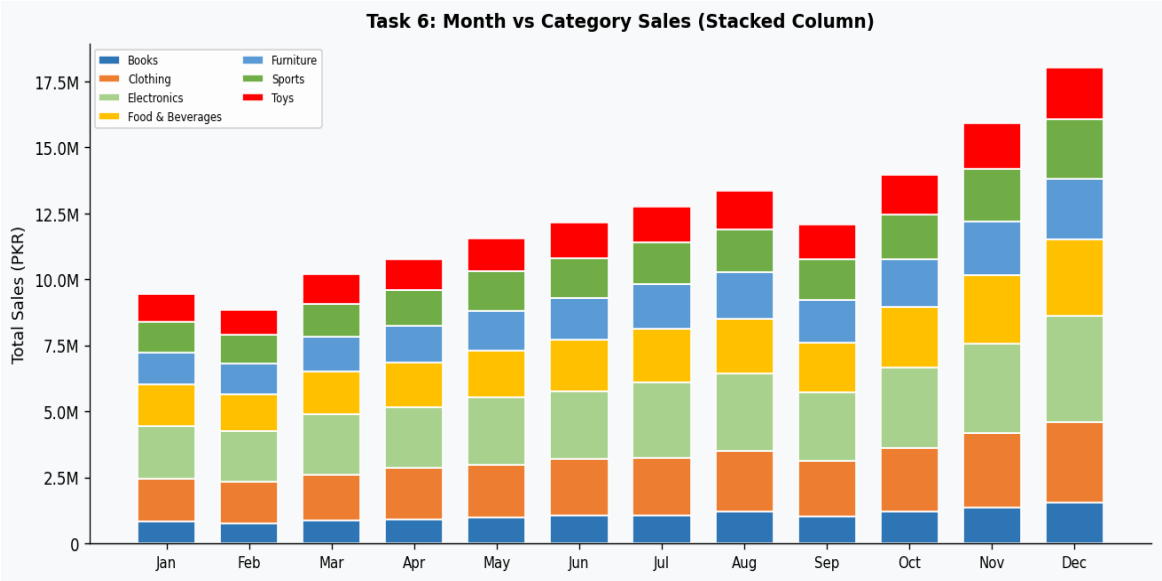
SQL Query

```
SELECT MONTH, CATEGORY,
       SUM(SALES) AS TOTAL_SALES
FROM   RETAIL_TRANSACTIONS
GROUP  BY MONTH, CATEGORY
ORDER  BY MONTH, CATEGORY;
```

Query Results

Month	Category	Total Sales
Jan	Books	PKR 826.2K
Jan	Clothing	PKR 1.62M
Jan	Electronics	PKR 2.01M
Jan	Food & Beverages	PKR 1.55M
Jan	Furniture	PKR 1.22M
Jan	Sports	PKR 1.17M
Jan	Toys	PKR 1.03M

Visualization



Task 7: Sales vs Units by Category

Compare total sales revenue and units sold side-by-side for each category to understand volume-value relationships.

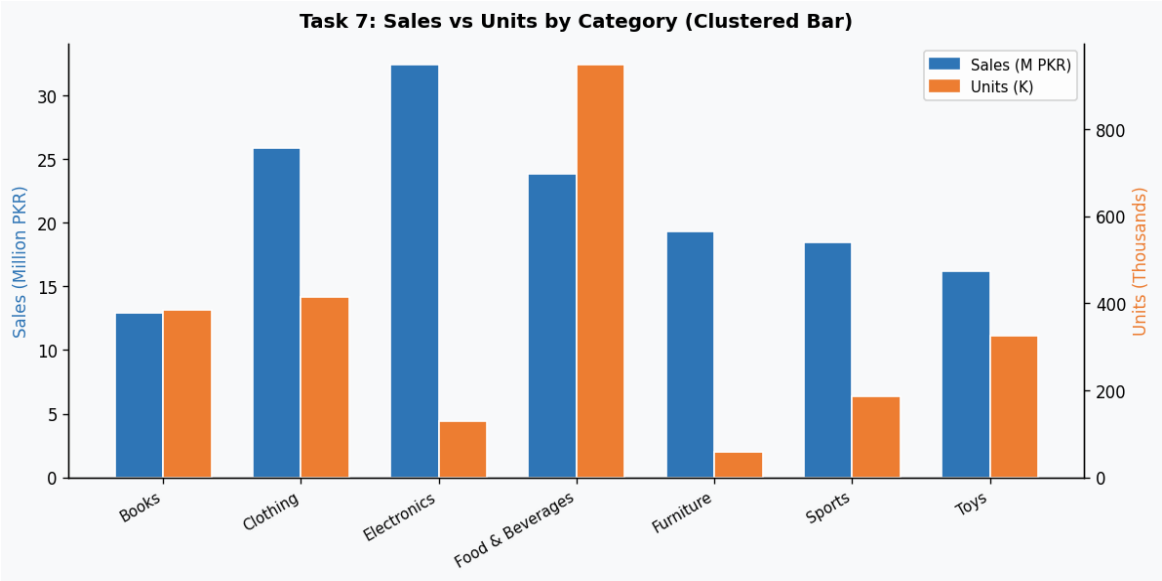
SQL Query

```
SELECT CATEGORY,
       SUM(SALES)  AS TOTAL_SALES,
       SUM(UNITS)  AS TOTAL_UNITS
FROM   RETAIL_TRANSACTIONS
GROUP  BY CATEGORY
ORDER  BY TOTAL_SALES DESC;
```

Query Results

Category	Total Sales	Total Units
Books	PKR 12.89M	385,371
Clothing	PKR 25.83M	414,085
Electronics	PKR 32.46M	129,578
Food & Beverages	PKR 23.84M	948,909
Furniture	PKR 19.34M	57,869
Sports	PKR 18.48M	185,184
Toys	PKR 16.20M	325,963

Visualization



Task 8: Monthly Sales by Region (Clustered)

Show monthly sales for each region to compare regional growth trajectories throughout the year.

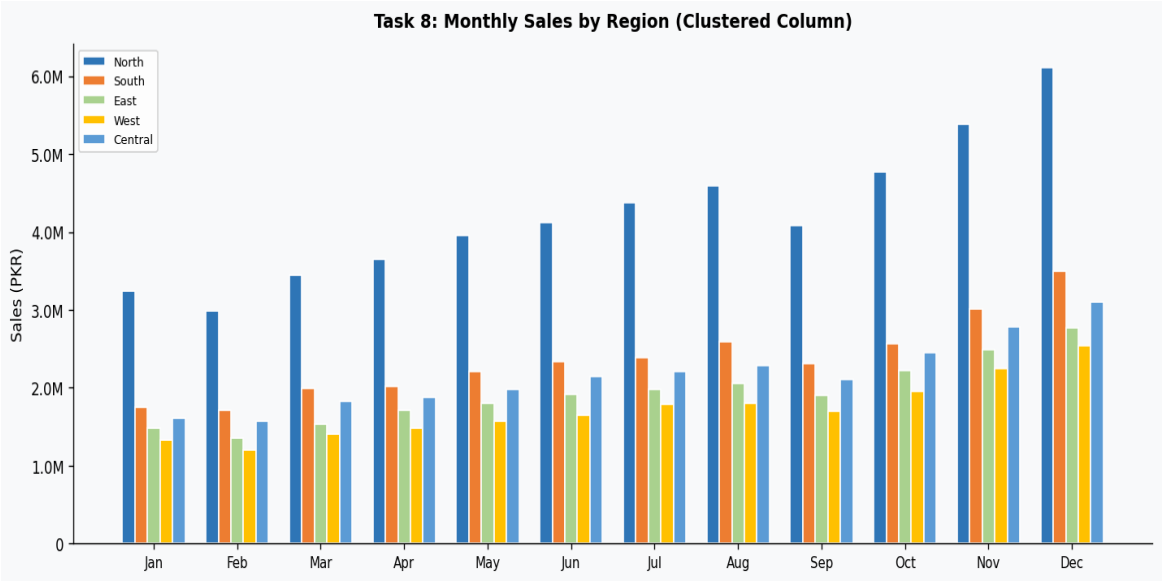
SQL Query

```
SELECT MONTH, REGION,
       SUM(SALES) AS TOTAL_SALES
FROM   RETAIL_TRANSACTIONS
GROUP  BY MONTH, REGION
ORDER  BY MONTH, REGION;
```

Query Results

Month	Region	Total Sales
Jan	Central	PKR 1.61M
Jan	East	PKR 1.49M
Jan	North	PKR 3.25M
Jan	South	PKR 1.76M
Jan	West	PKR 1.33M
Feb	Central	PKR 1.57M
Feb	East	PKR 1.36M
Feb	North	PKR 2.99M

Visualization



Task 9: Combo Chart (Sales + Units)

Display both sales revenue and units sold per month together in a combo chart to reveal decoupling patterns.

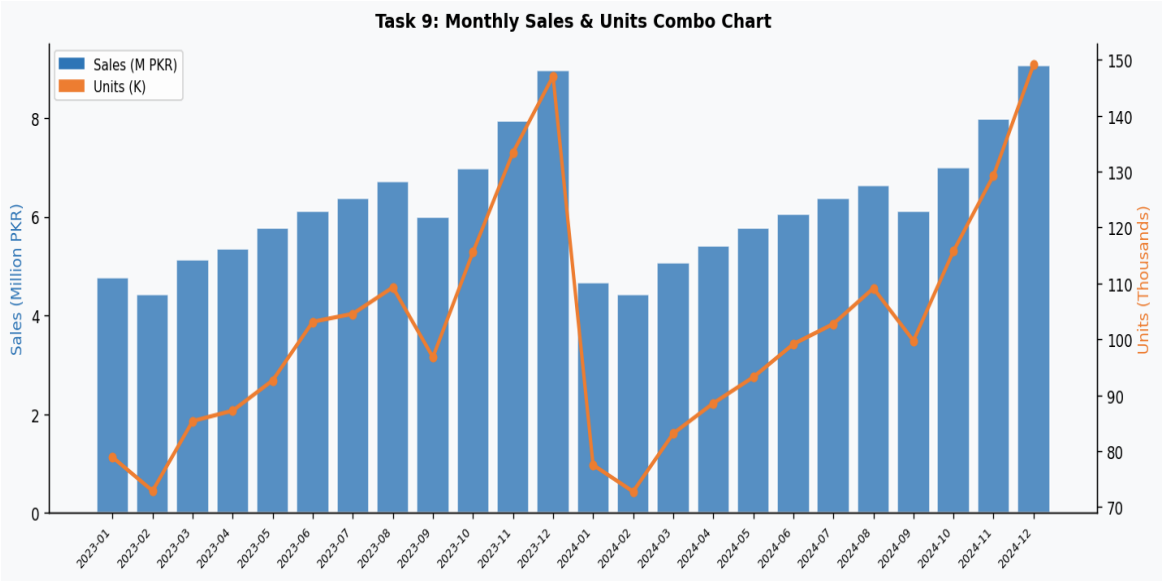
SQL Query

```
SELECT YEAR, MONTH,
       SUM(SALES) AS TOTAL_SALES,
       SUM(UNITS) AS TOTAL_UNITS
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, MONTH
ORDER  BY YEAR, MONTH;
```

Query Results

Year-Month	Total Sales	Total Units
2023-01	PKR 4.77M	78,899
2023-02	PKR 4.43M	72,893
2023-03	PKR 5.14M	85,363
2023-04	PKR 5.35M	87,227
2023-05	PKR 5.77M	92,660
2023-06	PKR 6.10M	103,123

Visualization



Task 10: Province-Level Sales Trend (Sindh)

Filter and visualize monthly sales specifically for Sindh province to track province-level performance over time.

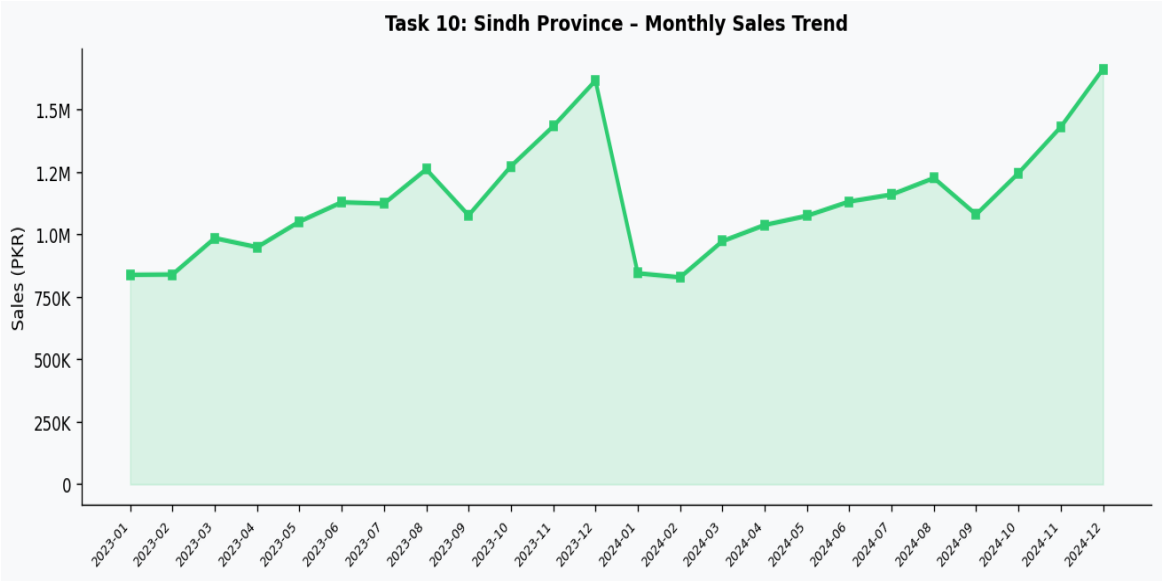
SQL Query

```
SELECT YEAR, MONTH,
       SUM(SALES) AS TOTAL_SALES
FROM   RETAIL_TRANSACTIONS
WHERE  STATE_PROVINCE = 'Sindh'
GROUP  BY YEAR, MONTH
ORDER  BY YEAR, MONTH;
```

Query Results

Year-Month	Sindh Sales
2023-01	PKR 838.5K
2023-02	PKR 839.9K
2023-03	PKR 985.5K
2023-04	PKR 949.2K
2023-05	PKR 1.05M
2023-06	PKR 1.13M

Visualization



Tasks 11–20: Advanced Analytics

Task 11: Region-wise Sales Contribution

Calculate each region's share of total company sales to understand contribution and geographic concentration.

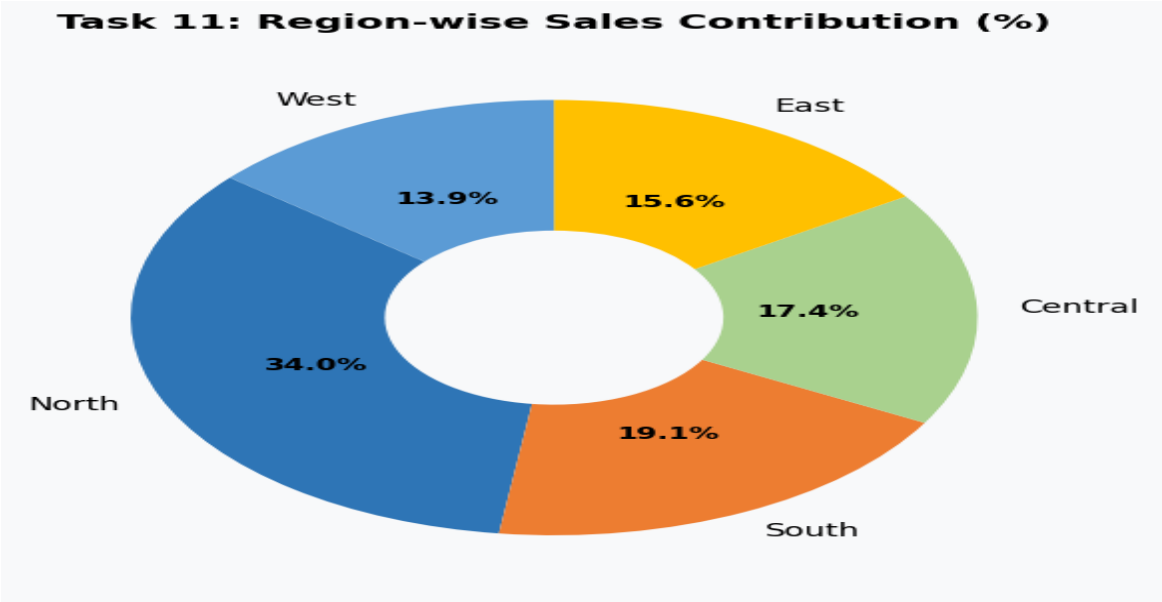
SQL Query

```
SELECT REGION,
       SUM(SALES)                                        AS TOTAL_SALES,
       ROUND(SUM(SALES) * 100.0 / SUM(SUM(SALES))
             OVER (), 2)                                 AS PCT_CONTRIBUTION
FROM   RETAIL_TRANSACTIONS
GROUP  BY REGION
ORDER  BY TOTAL_SALES DESC;
```

Query Results

Region	Total Sales	% Contribution
North	PKR 50.73M	34.03%
South	PKR 28.40M	19.05%
Central	PKR 25.98M	17.43%
East	PKR 23.26M	15.61%
West	PKR 20.68M	13.88%

Visualization



Task 12: Top Categories by Sales Performance

Rank all product categories by total sales and retrieve the top five, including their unit volumes.

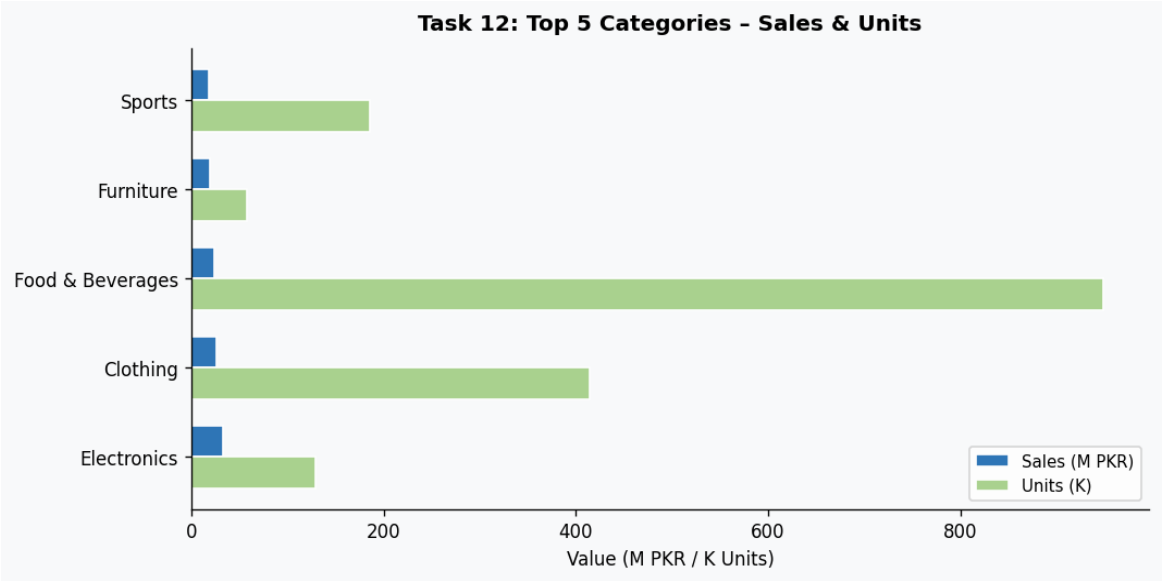
SQL Query

```
SELECT   CATEGORY,
         SUM(SALES) AS TOTAL_SALES,
         SUM(UNITS) AS TOTAL_UNITS
FROM     RETAIL_TRANSACTIONS
GROUP    BY CATEGORY
ORDER    BY TOTAL_SALES DESC
LIMIT    5;
```

Query Results

Rank	Category	Total Sales	Total Units
1	Electronics	PKR 32.46M	129,578
2	Clothing	PKR 25.83M	414,085
3	Food & Beverages	PKR 23.84M	948,909
4	Furniture	PKR 19.34M	57,869
5	Sports	PKR 18.48M	185,184

Visualization



Task 13: Monthly Sales Trend with Rolling Analysis

Compute monthly sales alongside a 3-month rolling average to smooth short-term fluctuations and reveal the true trend.

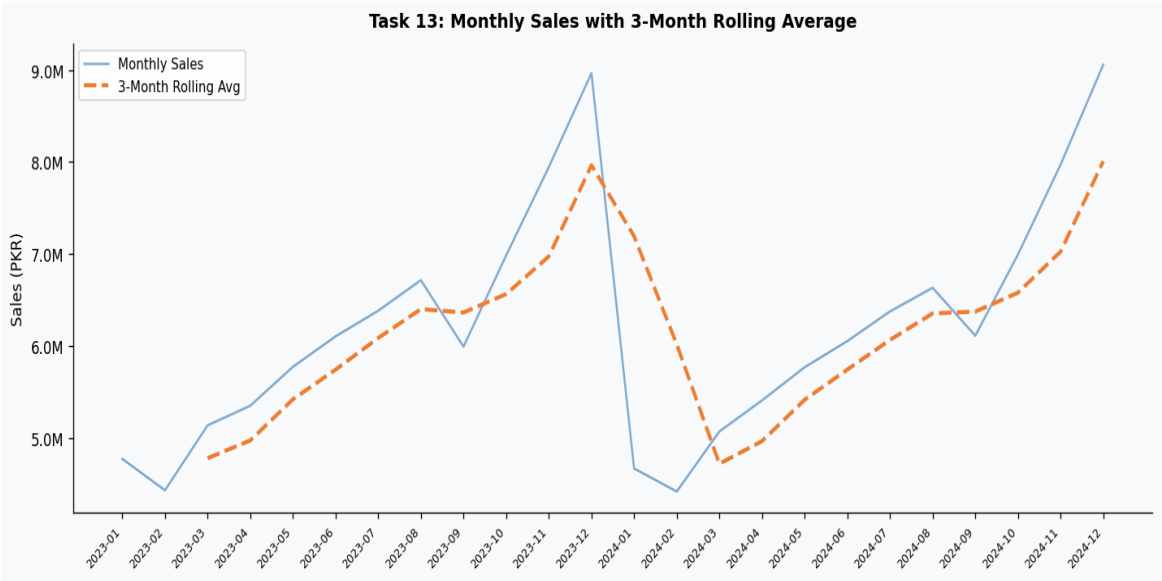
SQL Query

```
SELECT YEAR, MONTH,
       SUM(SALES)                              AS TOTAL_SALES,
       ROUND(AVG(SUM(SALES))
             OVER (ORDER BY YEAR, MONTH
                   ROWS BETWEEN 2 PRECEDING
                        AND CURRENT ROW), 2)   AS ROLLING_AVG_3M
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, MONTH
ORDER  BY YEAR, MONTH;
```

Query Results

Year-Month	Total Sales	3-Month Rolling Avg
2023-01	PKR 4.77M	N/A
2023-02	PKR 4.43M	N/A
2023-03	PKR 5.14M	PKR 4.78M
2023-04	PKR 5.35M	PKR 4.97M
2023-05	PKR 5.77M	PKR 5.42M
2023-06	PKR 6.10M	PKR 5.74M

Visualization



Task 14: Monthly Units Trend and Change Analysis

Show monthly unit volumes alongside the month-over-month change to quickly spot accelerating or decelerating demand.

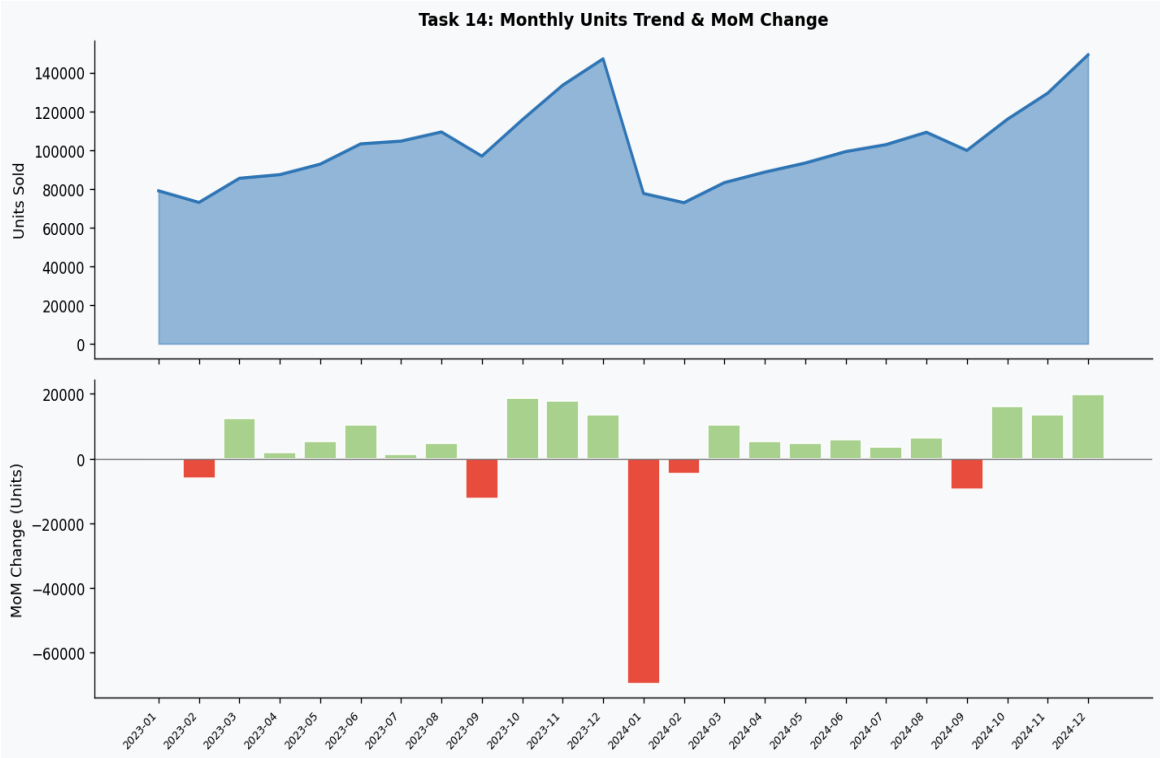
SQL Query

```
SELECT YEAR, MONTH,
       SUM(UNITS)                                                AS TOTAL_UNITS,
       SUM(UNITS) - LAG(SUM(UNITS))
                      OVER (ORDER BY YEAR, MONTH)                AS MOM_CHANGE
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, MONTH
ORDER  BY YEAR, MONTH;
```

Query Results

Year-Month	Total Units	MoM Change
2023-01	78,899	0
2023-02	72,893	-6,006
2023-03	85,363	12,470
2023-04	87,227	1,864
2023-05	92,660	5,433
2023-06	103,123	10,463

Visualization



Task 15: Regional Category Dominance Analysis

Use window functions to rank categories within each region and extract the top two, revealing regional category dominance.

SQL Query

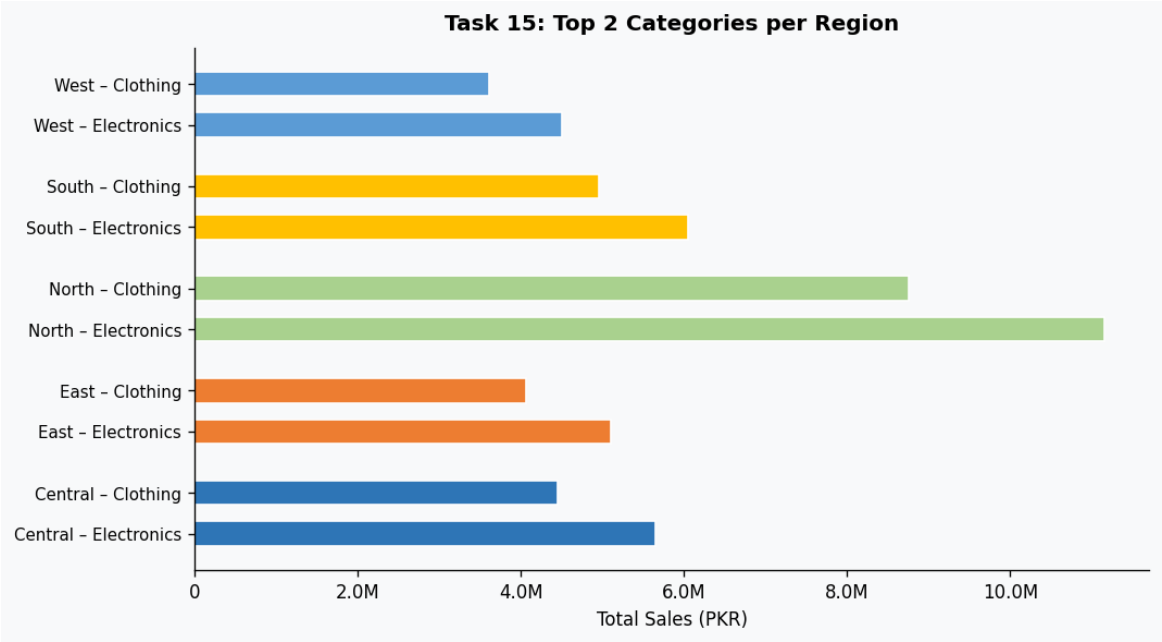
```
WITH RankedCategories AS (
  SELECT REGION, CATEGORY,
         SUM(SALES)                                          AS TOTAL_SALES,
         DENSE_RANK() OVER (PARTITION BY REGION
                            ORDER BY SUM(SALES) DESC)       AS SALES_RANK
  FROM   RETAIL_TRANSACTIONS
  GROUP  BY REGION, CATEGORY
)
SELECT REGION, CATEGORY, TOTAL_SALES, SALES_RANK
FROM   RankedCategories
WHERE  SALES_RANK <= 2
ORDER  BY REGION, SALES_RANK;
```

Query Results

Region	Category	Total Sales	Rank
Central	Electronics	PKR 5.65M	1
Central	Clothing	PKR 4.45M	2

East	Electronics	PKR 5.10M	1
East	Clothing	PKR 4.07M	2
North	Electronics	PKR 11.15M	1
North	Clothing	PKR 8.75M	2
South	Electronics	PKR 6.06M	1
South	Clothing	PKR 4.95M	2
West	Electronics	PKR 4.50M	1
West	Clothing	PKR 3.61M	2

Visualization



Task 16: Month vs Category Sales (Latest Year)

Restrict analysis to the most recent year (2024) to show how each category's monthly contribution evolved.

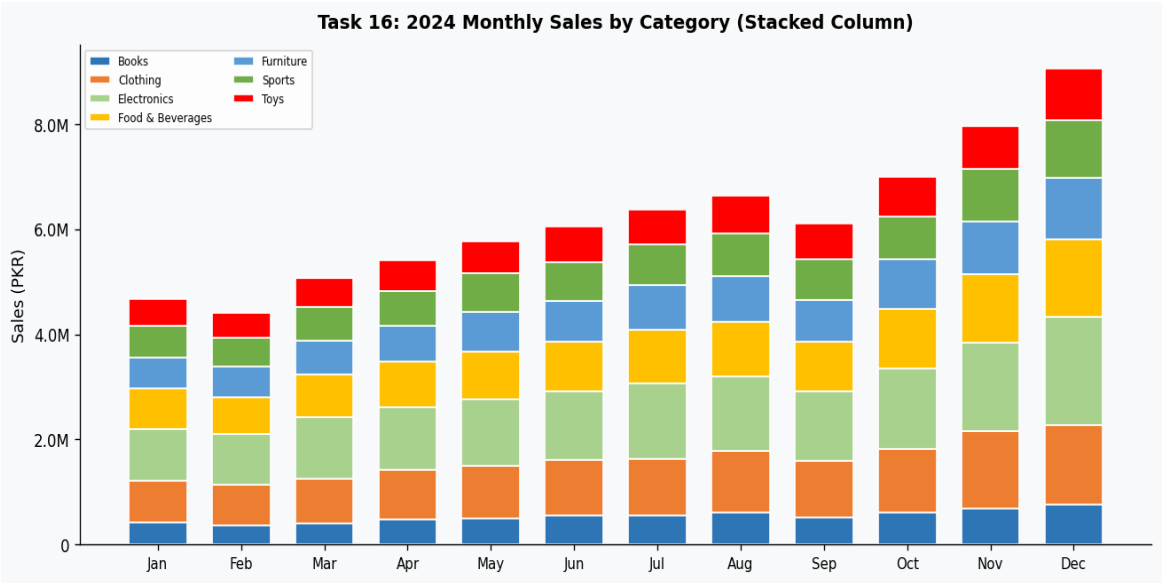
SQL Query

```
SELECT MONTH, CATEGORY,
       SUM(SALES) AS TOTAL_SALES
FROM   RETAIL_TRANSACTIONS
WHERE  YEAR = (SELECT MAX(YEAR) FROM RETAIL_TRANSACTIONS)
GROUP  BY MONTH, CATEGORY
ORDER  BY MONTH, CATEGORY;
```

Query Results

Month	Category	Total Sales (2024)
Jan	Books	PKR 415.4K
Jan	Clothing	PKR 795.0K
Jan	Electronics	PKR 985.6K
Jan	Food & Beverages	PKR 769.6K
Jan	Furniture	PKR 595.3K
Jan	Sports	PKR 594.5K
Jan	Toys	PKR 511.7K

Visualization



Task 17: Category-wise Average Selling Price Analysis

Compute Average Selling Price (ASP = Sales / Units) per category with NULLIF to safely handle zero-unit edge cases.

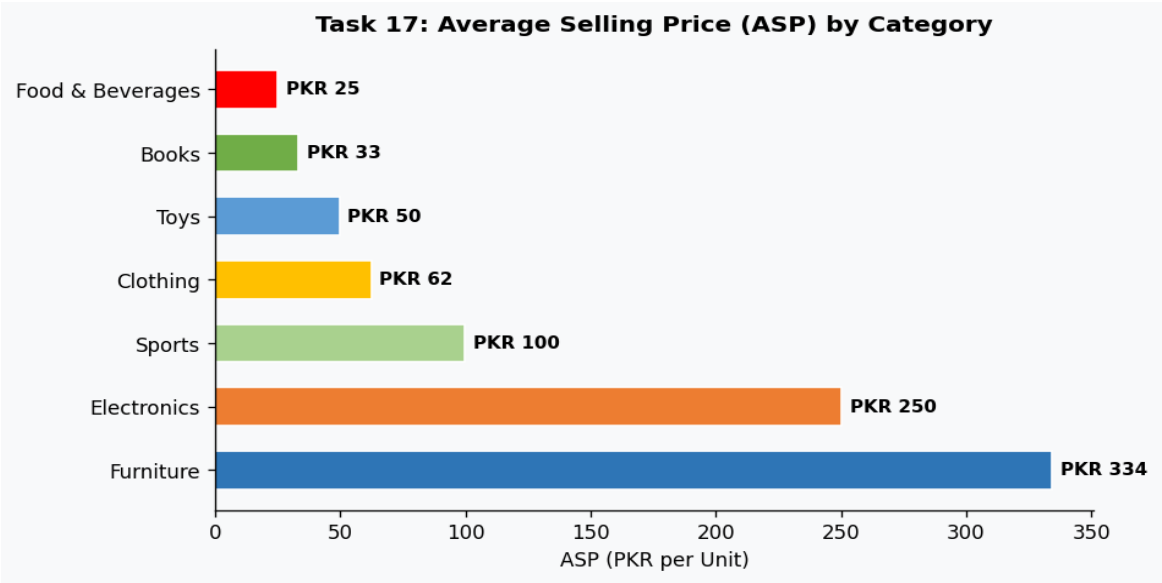
SQL Query

```
SELECT CATEGORY,
       SUM(SALES)                                        AS TOTAL_SALES,
       SUM(UNITS)                                        AS TOTAL_UNITS,
       ROUND(SUM(SALES) /
             NULLIF(SUM(UNITS), 0), 2)                  AS AVG_SELLING_PRICE
FROM   RETAIL_TRANSACTIONS
GROUP  BY CATEGORY
ORDER  BY AVG_SELLING_PRICE DESC;
```

Query Results

Category	Total Sales	Total Units	ASP
Books	PKR 12.89M	385,371	PKR 33.45
Clothing	PKR 25.83M	414,085	PKR 62.39
Electronics	PKR 32.46M	129,578	PKR 250.48
Food & Beverages	PKR 23.84M	948,909	PKR 25.13
Furniture	PKR 19.34M	57,869	PKR 334.26
Sports	PKR 18.48M	185,184	PKR 99.79
Toys	PKR 16.20M	325,963	PKR 49.71

Visualization



Task 18: Month vs Region Sales (Top Regions)

Limit the monthly regional view to only the top 3 regions by overall sales for a cleaner, focused comparison.

SQL Query

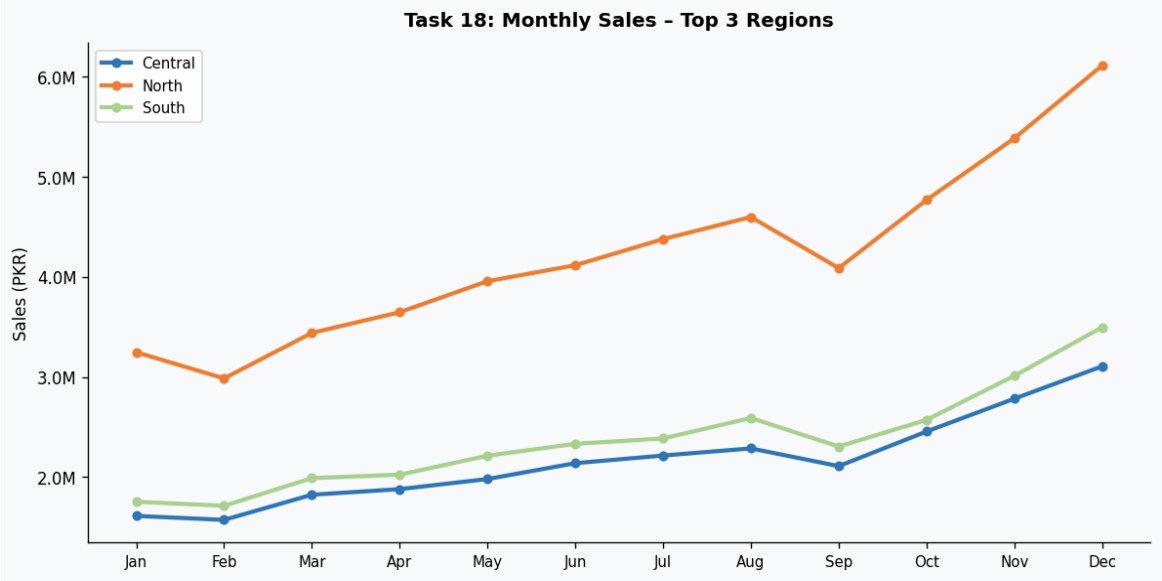
```
WITH TopRegions AS (
  SELECT REGION
  FROM   RETAIL_TRANSACTIONS
  GROUP  BY REGION
  ORDER  BY SUM(SALES) DESC
  LIMIT  3
)
SELECT t.MONTH, t.REGION,
       SUM(t.SALES) AS TOTAL_SALES
FROM   RETAIL_TRANSACTIONS t
```

```
JOIN   TopRegions tr ON t.REGION = tr.REGION
GROUP  BY t.MONTH, t.REGION
ORDER  BY t.MONTH, t.REGION;
```

Query Results

Month	Region	Total Sales
Jan	Central	PKR 1.61M
Jan	North	PKR 3.25M
Jan	South	PKR 1.76M
Feb	Central	PKR 1.57M
Feb	North	PKR 2.99M
Feb	South	PKR 1.71M
Mar	Central	PKR 1.82M
Mar	North	PKR 3.44M
Mar	South	PKR 1.99M

Visualization



Task 19: Monthly Sales, Units, and Growth Analysis

Combine sales, units, and month-over-month growth percentage in a single query for holistic performance monitoring.

SQL Query

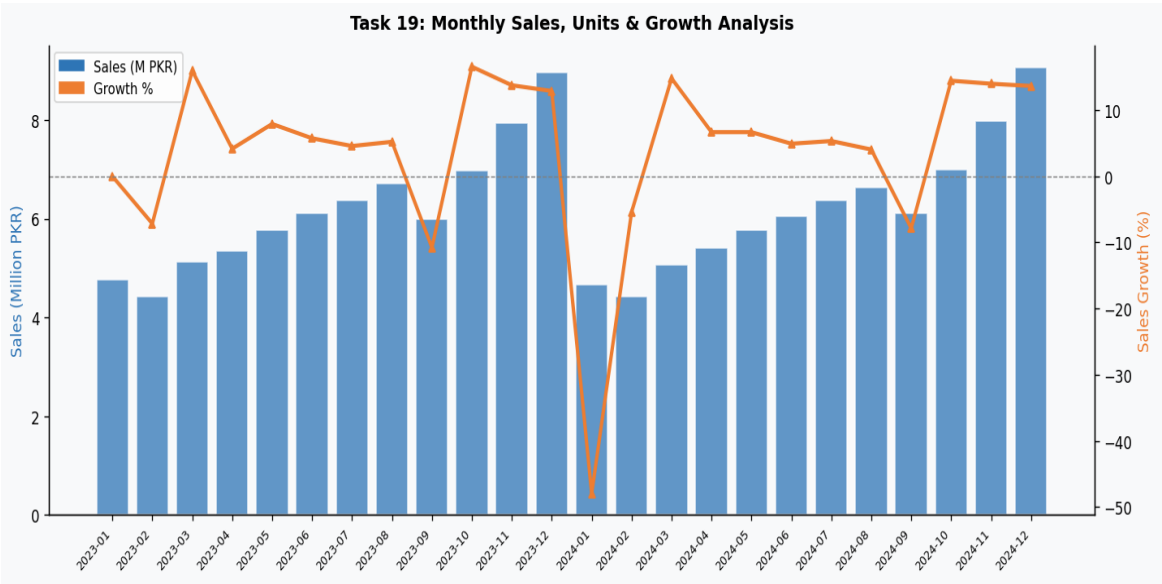
```
SELECT YEAR, MONTH,
       SUM(SALES)                                          AS TOTAL_SALES,
```

```
        SUM(UNITS)                                                  AS TOTAL_UNITS,
        ROUND((SUM(SALES) - LAG(SUM(SALES))
                OVER (ORDER BY YEAR, MONTH))
              / NULLIF(LAG(SUM(SALES))
                OVER (ORDER BY YEAR, MONTH), 0) * 100, 2)          AS SALES_GROWTH_PCT
FROM    RETAIL_TRANSACTIONS
GROUP   BY YEAR, MONTH
ORDER   BY YEAR, MONTH;
```

Query Results

Year-Month	Sales	Units	Growth %
2023-01	PKR 4.77M	78,899	0%
2023-02	PKR 4.43M	72,893	-7.16%
2023-03	PKR 5.14M	85,363	15.95%
2023-04	PKR 5.35M	87,227	4.16%
2023-05	PKR 5.77M	92,660	7.91%
2023-06	PKR 6.10M	103,123	5.74%

Visualization



Task 20: Province-wise Monthly Sales Ranking Analysis

For Sindh province, show monthly sales alongside its ranking among all provinces for every month using DENSE_RANK().

SQL Query

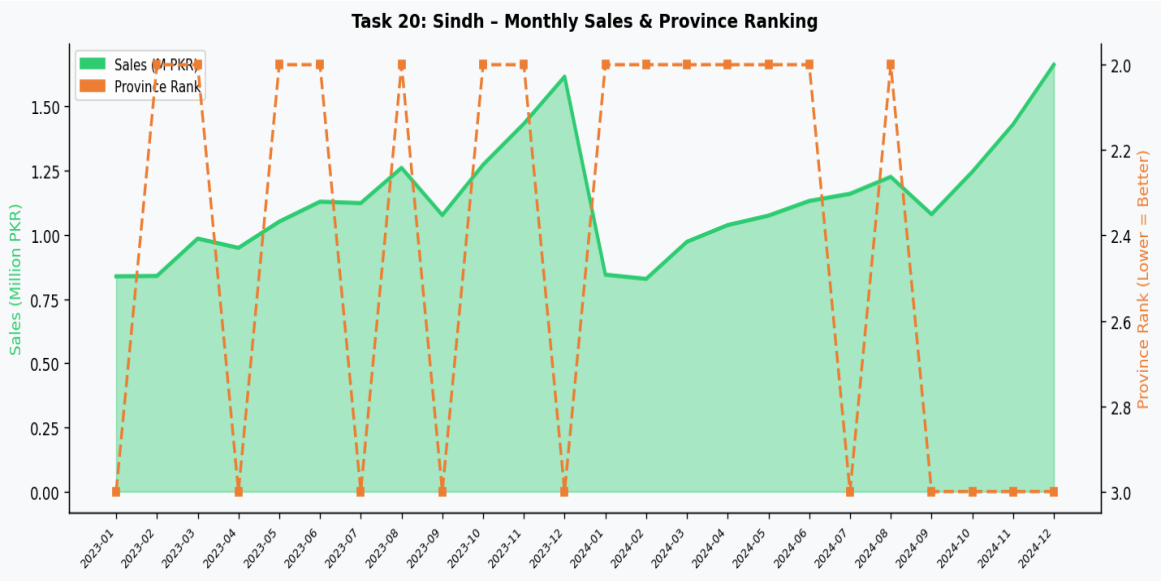
```
WITH ProvinceMonthly AS (
```

```
  SELECT YEAR, MONTH, STATE_PROVINCE,
         SUM(SALES)                                          AS TOTAL_SALES
  FROM   RETAIL_TRANSACTIONS
  GROUP  BY YEAR, MONTH, STATE_PROVINCE
),
Ranked AS (
  SELECT *,
         DENSE_RANK() OVER (PARTITION BY YEAR, MONTH
                            ORDER BY TOTAL_SALES DESC)      AS PROVINCE_RANK
  FROM   ProvinceMonthly
)
SELECT YEAR, MONTH, STATE_PROVINCE,
       TOTAL_SALES, PROVINCE_RANK
FROM   Ranked
WHERE  STATE_PROVINCE = 'Sindh'
ORDER  BY YEAR, MONTH;
```

Query Results

Year-Month	Sindh Sales	Province Rank
2023-01	PKR 838.5K	3
2023-02	PKR 839.9K	2
2023-03	PKR 985.5K	2
2023-04	PKR 949.2K	3
2023-05	PKR 1.05M	2
2023-06	PKR 1.13M	2
2023-07	PKR 1.12M	3
2023-08	PKR 1.26M	2

Visualization


Task 20: Sindh - Monthly Sales & Province Ranking
Sales (M-PKR)
Province Rank
Sales (Million PKR)
Province Rank (Lower = Better)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
2.0
2.2
2.4
2.6
2.8
3.0
2023-01
2023-02
2023-03
2023-04
2023-05
2023-06
2023-07
2023-08
2023-09
2023-10
2023-11
2023-12
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07
2024-08
2024-09
2024-10
2024-11
2024-12